

Adham Mohamed Hassan

ml_engineering | 2.0 years

About Me

Junior Machine Learning Engineer with 2 years of experience building and deploying ML models in fintech, healthtech, and developer tools. Graduate of Zewail City of Science and Technology with deep expertise in deep learning, NLP, and MLOps. Active open-source contributor to TensorFlow, PyTorch, and Hugging Face Transformers. Eager to deliver robust, scalable AI solutions in collaborative, fast-paced environments.

Career History

AI Intern — IBM Egypt

2021-06 to 2021-09

Developed NLP text classification models for a customer service chatbot using Python and scikit-learn. Preprocessed Arabic-English bilingual data, implemented TF-IDF and word embeddings, and achieved 92% accuracy on intent classification tasks.

Machine Learning Research Assistant — Zewail City of Science and Technology

2021-09 to 2022-06

Assisted faculty research on attention mechanisms for Arabic NLP. Built and evaluated Transformer variants in PyTorch, contributing to a paper on low-resource language modeling. Ran experiments on GPU clusters and maintained reproducible codebases.

Junior Data Scientist — Fawry

2022-07 to 2022-12

Designed and deployed fraud detection models for digital payments, reducing false positives by 15%. Built feature engineering pipelines with Pandas and SQL, and served models via REST APIs using FastAPI and Docker.

Machine Learning Engineer — Vezeeta

2023-01 to 2023-07

Developed a doctor recommendation engine using two-tower neural networks, improving patient booking conversion by 12%. Implemented NLP modules with Hugging Face Transformers for symptom understanding and deployed models using FastAPI and Docker Compose.

Junior Machine Learning Engineer — Instabug

2023-08 to Present

Building automated bug detection and crash analysis models for mobile apps. Fine-tune Large Language Models for log summarization and anomaly detection. Optimize inference pipelines with ONNX and Triton Inference Server, and contribute to internal TensorFlow and PyTorch libraries.

Academic Background

Bachelor of Science in Computer Engineering, Computer Engineering — Zewail City of Science and Technology (2022-06)

Expertise

Python · C++ · TensorFlow · PyTorch · Hugging Face Transformers · scikit-learn · Docker · FastAPI · SQL · Pandas · NumPy · ONNX · Triton Inference Server · Git · Linux · Arabic · English (Professional)

Credentials

TensorFlow Developer Certificate (Google)
AWS Certified Machine Learning - Specialty (Amazon Web Services)
Deep Learning Specialization (deeplearning.ai (Coursera))

Side Projects

system-design-primer — Learn how to design large-scale systems. Prep for the system design interview. Includes Anki flashcards.

- | **awesome-python** — An opinionated list of Python frameworks, libraries, tools, and resources
- | **project-based-learning** — Curated list of project-based tutorials
- | **Python** — All Algorithms implemented in Python
- | **tensorflow** — An Open Source Machine Learning Framework for Everyone